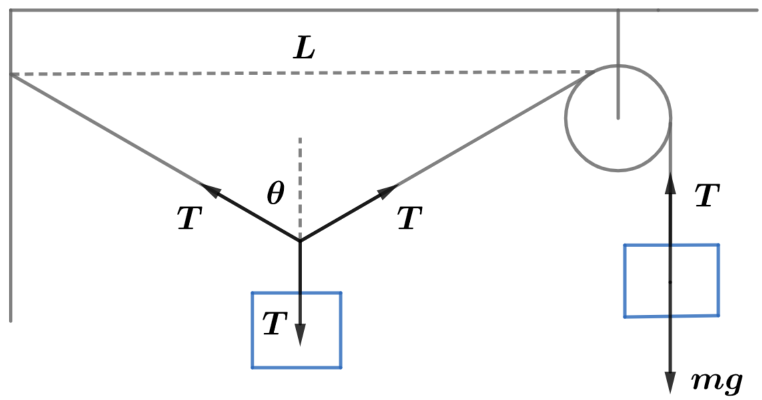
2020A F=ma Exam: Problem 10

Kevin S. Huang



Since the tension is the same throughout a massless string and the left mass is at rest, we have $\theta = \pi/3$ by symmetry. The change in the length of the string left of the pulley is

$$\Delta L = \frac{L}{\sin\theta} - L = \left(\frac{2}{\sqrt{3}} - 1\right) L$$

By conservation of string,

$$\Delta h = \Delta L = 0.15L$$

so the answer is $\boxed{\text{A}}$.